

Configuring a DHCP Option Set in AWS VPC

To Begin with the Lab

Summary of the Lab

This lab demonstrated how to configure and apply a custom DHCP Option Set in an AWS VPC. A Windows EC2 instance was launched, connected via RDP, and its default DHCP settings were verified. A new DHCP Option Set with a custom domain name and DNS server was created, associated with the VPC, and validated by renewing the DHCP lease and confirming the updated network configuration.

- Sign in to the AWS Management Console.
- Navigate to EC2.
- Click Launch Instance.
- Enter an instance name (for example, Windows-Server).
- Select a Windows Server AMI

EC2 > Instances > Launch an instance

Name and tags [Info](#)

Name: cloud_freeks_ec2 [Add additional tags](#)

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

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Recents My AMIs **Quick Start**

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux Debian

Amazon Machine Image (AMI)

Microsoft Windows Server 2025 Base [Free tier eligible](#)

ami-096294801130d9196 (64-bit (x86))
Virtualization: hvm ENA enabled: true Root device type: ebs

Description
Microsoft Windows 2025 Datacenter edition. [English]

Summary

[Cancel](#) [Launch instance](#) [Preview code](#)

- Choose an instance type (t3.micro).
- Select or create a key pair.

Instance type [Info](#) [Get advice](#)

Instance type

t3.micro [Free tier eligible](#)

Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0104 USD per Hour
On-Demand Windows base pricing: 0.0196 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour
On-Demand SUSE base pricing: 0.0104 USD per Hour On-Demand RHEL base pricing: 0.0392 USD per Hour

[Additional costs apply for AMIs with pre-installed software](#)

Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

jenkins [Create new key pair](#)

For Windows instances, you use a key pair to decrypt the administrator password. You then use the decrypted password to connect to your instance.

- Under Network Settings:
- Select the desired VPC.

- ▼ Network settings [Info](#)

VPC - *required* | [Info](#)

vpc-099792461c7d0a616
172.31.0.0/16

(default) ▼

↻

Subnet | [Info](#)

subnet-0b5783ef03796c671
VPC: vpc-099792461c7d0a616 Owner: 463646775279
Availability Zone: us-east-1a (use1-az4) Zone type: Availability Zone
IP addresses available: 4090 CIDR: 172.31.16.0/20

▼

↻ Create new subnet [↗](#)

Auto-assign public IP | [Info](#)

Enable ▼

Additional charges apply when outside of [free tier allowance](#)

- Firewall (security groups) | Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

Security group name - required

dhcp-sg

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _~/[]@!*=&()|!\$*

Description - required | Info

dhcp-sg

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 3389, 0.0.0.0/0)

Remove

Type Info	Protocol Info	Port range Info
rdp	TCP	3389
Source type Info	Source Info	Description - optional Info
Anywhere	Q Add CIDR, prefix list or security group 0.0.0.0/0 ✕	e.g. SSH for admin desktop

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Add security group rule

Summary

Cancel

Launch instance

Preview code

- EC2 > Instances > i-0b5041f6fc1995edb > Connect to instance

SSM Session Manager

RDP client

EC2 serial console

Instance ID

i-0b5041f6fc1995edb (cloud_freeeks_ec2)

Connection Type

Connect using RDP client

Download a file to use with your RDP client and retrieve your password.

Connect using Fleet Manager

To connect to the instance using Fleet Manager Remote Desktop, the SSM Agent must be installed and running on the instance. For more information, see [Working with SSM Agent](#)

You can connect to your Windows instance using a remote desktop client of your choice, and by downloading and running the RDP shortcut file below:

Download remote desktop file

When prompted, connect to your instance using the following username and password:

Public DNS

ec2-100-27-227-138.compute-1.amazonaws.com

Username info

Administrator

Password

Get password

If you've joined your instance to a directory, you can use your directory credentials to connect to your instance.

- Now click on the Get password
- Upload your private key to create the password
- Click Decrypt Password

Use your private key to retrieve and decrypt the initial Windows administrator password for this instance.

Instance ID
i-0b5041f6fc1995edb (cloud_freaks_ec2)

Key pair associated with this instance
jenkins

Private key
Either upload your private key file or copy and paste its contents into the field below.

[Upload private key file](#)

jenkins.pem
1.68 KB

Private key contents

```
-----BEGIN RSA PRIVATE KEY-----
MIIEpQIBAAKCAQEA2i37qVnytlWc9FWe8eFRpBz0W0J/y45IfqTH6RalbJeyskm
ITS0TTJauYnuXZcGRcygi75LJK5++sp32yvf9mW12wqUX9nGWMMapd5tcJzdailEQ
qTw6q+mB5Gg57TFklmHgU2DoP4cPAo0b4niHORw+SslGnN5G3bC/gdRTG2A3LbT9
g0PTdmjkawlfj19UpB4d52Ox9HRDhX1omd5FLOdbtbPg6i5nwlpmppGjweyrcLq3I
he+gie/fuKpauTiBAqvz7dksyl00ea7YqhNDn5nM4yuN56nLLun3Quccp+wJgHZAT7
keaqE8MYCHM3ZkKXYBgRqXmKvGqgw1UqwbC/KQIDAQABaoIBAQDITWT48jpP51jA
PlcC23A/cdGmdlB+uga50f2GQaM3kgGdisKZAYXrj36jrYGuCsE5TfRtO7M73
-----
```

[Cancel](#) [Decrypt password](#)

and you will get your instance password

Public DNS
ec2-100-27-227-138.compute-1.amazonaws.com

Username
Administrator

Password
XD3?F5(vUG.9fMJlcpj?xA7HMOX*qR?

If you've joined your instance to a directory, you can use your directory credentials to connect to your instance.

[Cancel](#)

- Retrieve the Windows administrator password and connect using Remote Desktop (RDP).
- Now open the Remote Desktop, if you are on windows OS
- Copy the public IP of ec2 instance and the user name and paste there

Instance ID
i-0b5041f6fc1995edb (cloud_freaks_ec2)

Connection Type
[Connect using RDP client](#)
Download a file to use with your RDP client and retrieve your password.

You can connect to your Windows instance using a remote desktop client of your choice.

[Download remote desktop file](#)

When prompted, connect to your instance using the following username and password:

Public DNS
ec2-100-27-227-138.compute-1.amazonaws.com

Password
XD3?F5(vUG.9fMJlcpj?xA7HMOX*qR?

If you've joined your instance to a directory, you can use your directory credentials to connect to your instance.

Remote Desktop Connection

General | Display | Local Resources | Experience | Advanced

Logon settings

Enter the name of the remote computer.

Computer: 100.27.227.138

User name: Administrator

You will be asked for credentials when you connect.

☐ Allow me to save credentials

Connection settings

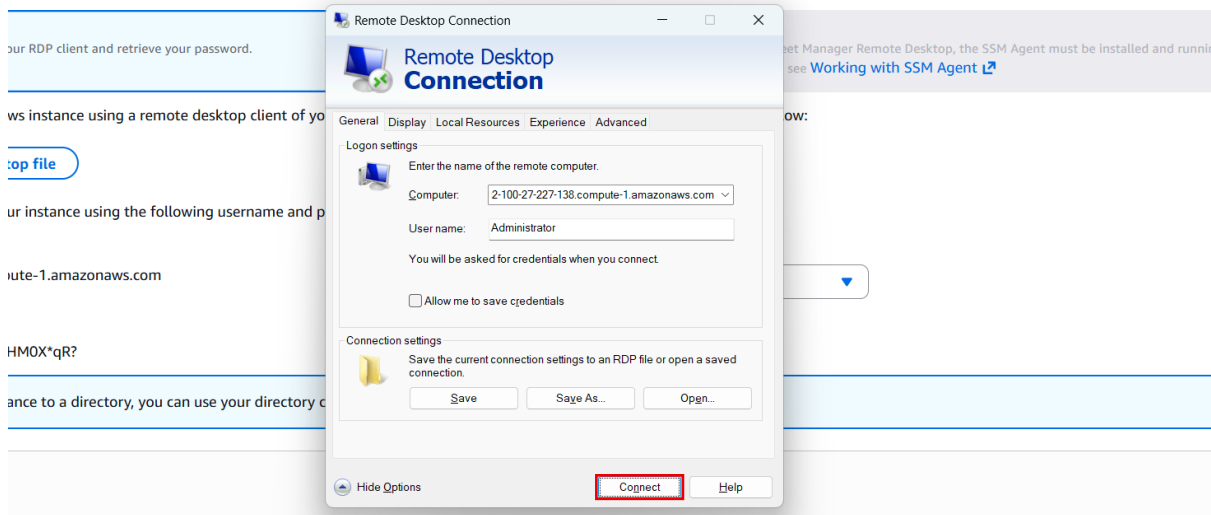
Save the current connection settings to an RDP file or open a saved connection.

[Save](#) [Save As...](#) [Open...](#)

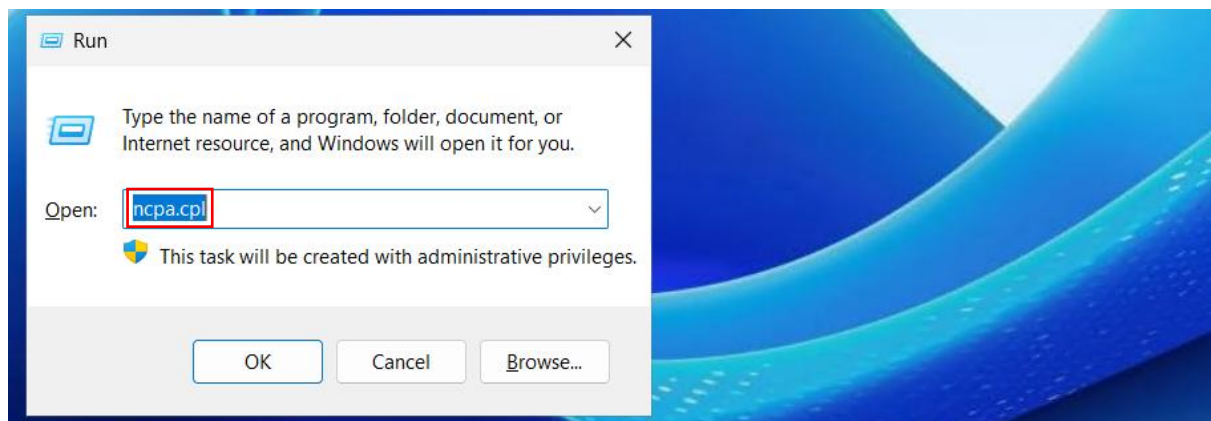
[Hide Options](#) [Connect](#) [Help](#)

[Cancel](#)

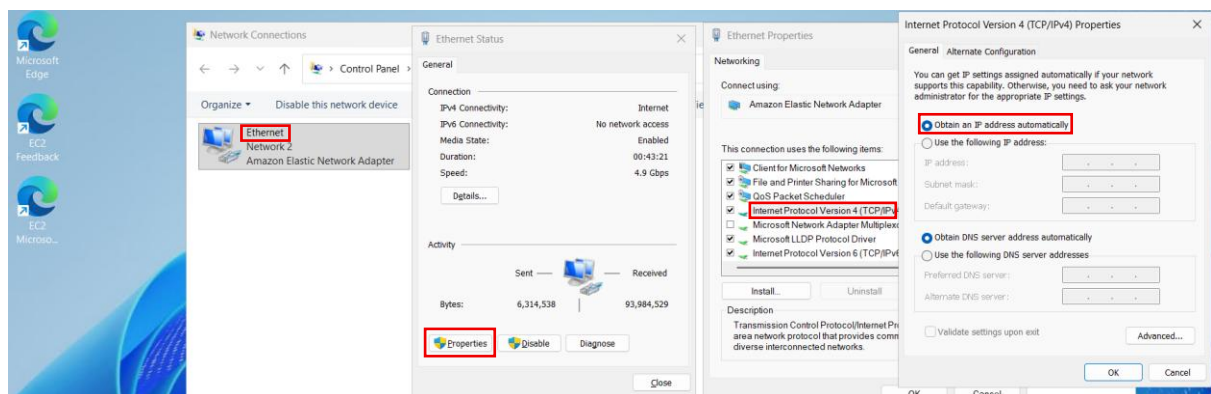
- Then click Open on Remote Desktop
- Upload the file the you have downloaded.
- Click Connect



- Now it will ask you to enter the password, copy the password that was generated and paste
- And you will be connected to the windows instance
- After connecting to the instance, press Win + R on the instance and type **ncpa.cpl**



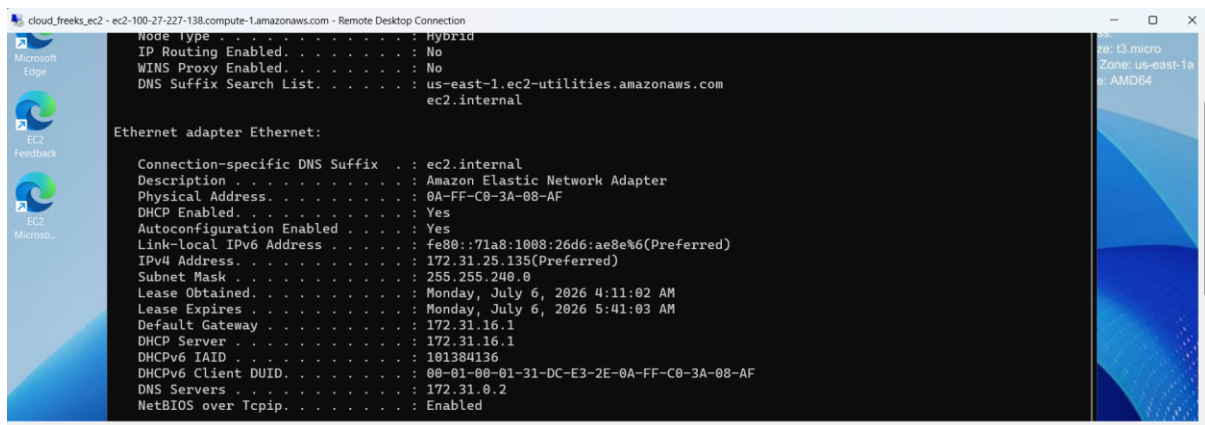
- Open **Network Connections** on the Windows instance.
- Navigate to the properties of the active network adapter and open the **IPv4 Properties**.
- Verify that both **Obtain an IP Address Automatically** and **Obtain DNS Server Address Automatically** are enabled.



- This confirms that the EC2 instance is receiving its network configuration through AWS DHCP.
- Open the **Command Prompt** on the EC2 instance.
- Run the following command to view the current network configuration:

```
ipconfig /all
```

- Verify the following information: Private IP Address, DNS Server, Default Gateway, DNS Suffix
- These values are currently being provided by the default AWS DHCP Option Set.



- Return to the **AWS Management Console** and navigate to the **VPC** service.
- From the left navigation pane, select **DHCP Option Sets**.
- Click **Create DHCP Option Set**.
- Provide a name for the DHCP Option Set
- Enter your preferred domain name, for example:
 - cloudfreeks.local
- This domain name will be assigned to all EC2 instances associated with this DHCP Option Set.
- Specify the DNS server IP address.
- For demonstration purposes, use:
 - 8.8.8.8
- In production environments, this would typically be the IP address of your own DNS or Active Directory server.

Create DHCP option set [Info](#)

Dynamic Host Configuration Protocol (DHCP) provides a standard for passing configuration information to hosts on a TCP/IP network. The options field of a DHCP message contains configuration parameters.

Tag settings

DHCP option set name - optional

cloudfreeks-dhcp

DHCP option

Specify at least one configuration parameter.

Domain name [Info](#)

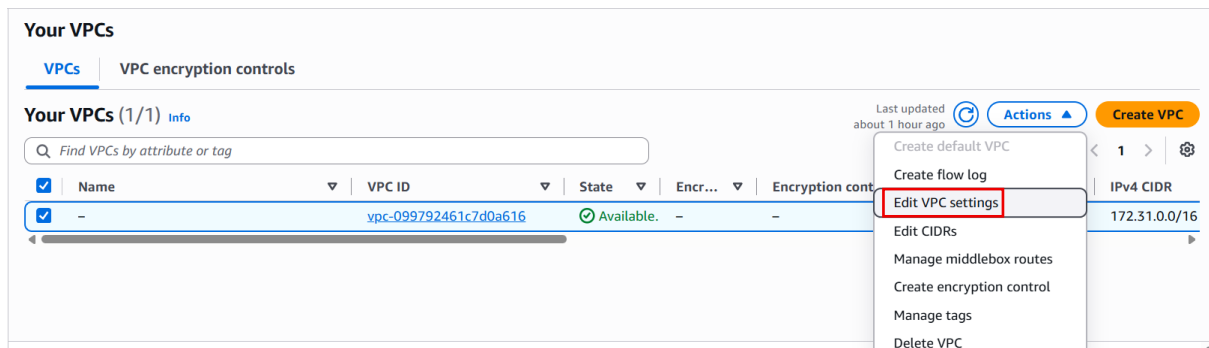
cloudfreeks.local

Domain name servers [Info](#)

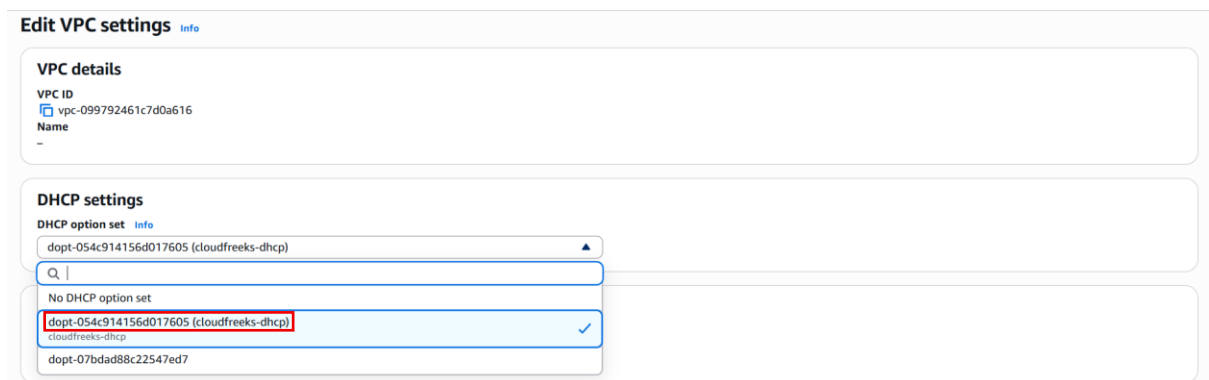
8.8.8.8

Enter up to four IPv4 addresses and four IPv6 addresses, separated by commas.

- Associate the DHCP Option Set with the VPC
- Navigate to the VPC section.
- Select the VPC where your EC2 instances are running.
- Click Actions → Edit VPC Settings.

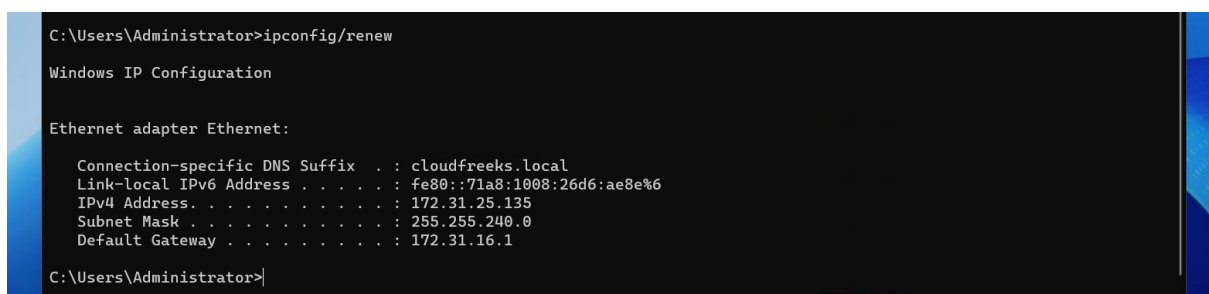


- Select the newly created DHCP Option Set.
- Save the changes



- Renew the DHCP Lease
- Return to the Windows EC2 instance.
- Open Command Prompt as Administrator.
- Run the following command:

```
ipconfig /renew
```



- Run the following command again:

```
ipconfig /all
```

- Verify that the following values have changed:
- DNS Suffix → cloudfox.local
- DNS Server → 8.8.8.8
- This confirms that the EC2 instance is now using the custom DHCP Option Set.

```

WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : us-east-1.ec2-utilities.amazonaws.com
                                  ec2.internal

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : cloudfreeks.local
    Description . . . . . : Amazon Elastic Network Adapter
    Physical Address. . . . . : 0A-FF-C0-3A-08-AF
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::71a8:1008:26d6:ae8e%6(Preferred)
    IPv4 Address. . . . . : 172.31.25.135(Preferred)
    Subnet Mask . . . . . : 255.255.240.0
    Lease Obtained. . . . . : Monday, July 6, 2026 4:11:02 AM
    Lease Expires . . . . . : Monday, July 6, 2026 7:11:06 AM
    Default Gateway . . . . . : 172.31.16.1
    DHCP Server . . . . . : 172.31.16.1
    DHCPv6 IAID . . . . . : 101384136
    DHCPv6 Client DUID. . . . . : 00-01-00-01-31-DC-E3-2E-0A-FF-C0-3A-08-AF
    DNS Servers . . . . . : 8.8.8.8
    NetBIOS over Tcpip. . . . . : Enabled

```